

PowerBI_Documentation_Explained(Sheet1)

Category	Measure Name	Description	DAX Expression
Climate	Heatwave Count	Counts the number of rows to calculate heatwave count.	COUNTROWS('Fact_Heatwaves')
Climate	Avg Monthly Heatwaves	Calculates the average value for avg monthly heatwaves over the selected context.	AVERAGEX(SUMMARIZE('Model Calendar'; 'Model Calendar'[Year]; 'Model Calendar'[Month_Num]); CALCULATE([Heatwave Count]))
General	Total Days in Period	Counts the number of rows to calculate total days in period.	COUNTROWS (VALUES ('Model Calendar'[Date]))
General	Annual Avg Anomaly	Calculates the average value for annual avg anomaly over the selected context.	AVERAGEX(VALUES('Model Calendar'[Year_Num]); CALCULATE(AVERAGE('Fact_Daily_Anamoly'[Temp Anomaly (°C)])))
Climate	Cumulative Warming	Returns the maximum value for cumulative warming within the current filter context.	CALCULATE([Annual Avg Anomaly]; FILTER(ALL('Model Calendar'[Date]); 'Model Calendar'[Date] <= MAX('Model Calendar'[Date])))
Climate	Rolling 30D Warming	Calculates the average value for rolling 30d warming over the selected context.	AVERAGEX(DATESINPERIOD('Model Calendar'[Date]; MAX('Model Calendar'[Date]); -30; DAY); [Annual Avg Anomaly])
Climate	Warming Rate	Calculates warming rate as a ratio between two values using safe division.	DIVIDE([Temp Monthly Anomaly (°C)] - CALCULATE([Temp Monthly Anomaly (°C)]; DATEADD('Model Calendar'[Date]; -1; YEAR)); 1)
Trend	Slope	Calculates the average value for slope over the selected context.	VAR TableYear = SUMMARIZE('Model Calendar'; 'Model Calendar'[Year_Num]; "Y"; CALCULATE(AVERAGE('Fact_Daily_Anamoly'[Temp Anomaly (°C)]))) VAR AvgX = AVERAGEX(TableYear; 'Model Calendar'[Year_Num]) VAR AvgY = AVERAGEX(TableYear; [Y]) RETURN DIVIDE(SUMX(TableYear; ('Model Calendar'[Year_Num] - AvgX) * ([Y] - AvgY)); SUMX(TableYear; ('Model Calendar'[Year_Num] - AvgX)^2))
General	Intercept	Calculates the average value for intercept over the selected context.	VAR AvgX = AVERAGE('Model Calendar'[Year_Num]) VAR AvgY = [Annual Avg Anomaly] RETURN AvgY - [Slope] * AvgX
Trend	Linear Trend	Computes linear trend based on the defined DAX calculation.	[Slope] * SELECTEDVALUE('Model Calendar'[Year_Num]) + [Intercept]
Trend	Trend per Decade	Computes trend per decade based on the defined DAX calculation.	[Slope] * 10
Precipitation	Avg Snowfall (cm)	Calculates the average value for avg snowfall (cm) over the selected context.	AVERAGE('Dim_monthly_precip'[Monthly_Average_Snow_cm])
Precipitation	Total dailyPrecipitation	Calculates total dailyprecipitation by summing the underlying values.	CALCULATE(SUM(Dim_Daily_precip_2015[Total Precip (mm)]))
Climate	Max Monthly Temperature (°C)	Returns the maximum value for max monthly temperature (°c) within the current filter context.	MAX (Dim_monthly_temperature[Monthly_Max_Temp_C])
Climate	Min Monthly Temperature (°C)	Returns the minimum value for min monthly temperature (°c) within the current filter context.	MIN (Dim_monthly_temperature[Monthly_Min_Temp_C])
Climate	Avg Monthly Temp	Calculates the average value for avg monthly temp over the selected context.	AVERAGE (Fact_mClimate[Monthly_Mean_Temp_C])
Climate	Temp Monthly Anomaly (°C)	Calculates the average value for temp monthly anomaly (°c) over the selected context.	[Avg Monthly Temp] - AVERAGE (Fact_mClimate[Baseline Mean Temp (°C)])
Climate	MoM Temperature Change (°C)	Calculates the change in mom temperature change (°c) compared to a previous time period.	[Avg Monthly Temp] - CALCULATE([Avg Monthly Temp]; DATEADD ('Model Calendar'[Date]; -1; MONTH))
Precipitation	Avg Monthly Precipitation (mm)	Calculates the average value for avg monthly precipitation (mm) over the selected context.	AVERAGE (Fact_mClimate[Monthly_Average_Precip_mm])
Wind	Avg Monthly Wind Gust (km/h)	Calculates the average value for avg monthly wind gust (km/h) over the selected context.	AVERAGE (Fact_mClimate[Monthly_Max_Gust_kmh])
AQI	AQI–Temperature Ratio	Calculates aqi–temperature ratio as a ratio between two values using safe division.	DIVIDE([Avg Station AQI]; [Avg Monthly Temp])
Precipitation	Total Precipitation	Calculates total precipitation by summing the underlying values.	SUM('Fact_mClimate'[Monthly_Average_Precip_mm])
Precipitation	Avg Rainfall (mm)	Calculates the average value for avg rainfall (mm) over the selected context.	AVERAGE('Fact_mClimate'[Monthly_Average_Rain_mm])
Precipitation	Rain to Precip Ratio	Calculates rain to precip ratio as a ratio between two values using safe division.	DIVIDE([Avg Rainfall (mm)]; [Avg Monthly Precipitation (mm)]; 0)
Climate	Avg Monthly Temp pred	Calculates the average value for avg monthly temp pred over the selected context.	AVERAGEX(VALUES(Fact_mClimate[Date]); CALCULATE(AVERAGE(Fact_mClimate[Monthly_Mean_Temp_C])))
Climate	Avg Temperature	Calculates the average value for avg temperature over the selected context.	CALCULATE(AVERAGE(Dim_Daily_temp__2015[Mean Temp (°C)]))
Climate	Corr Temp vs Pollution	Calculates corr temp vs pollution as a ratio between two values using safe division.	VAR MeanX = [Avg Temperature] VAR MeanY = [Avg Pollution] VAR T = SUMX(VALUES('Model Calendar'[Date]); ([Avg Temperature] - MeanX) * ([Avg Pollution] - MeanY)) VAR Denominator = SQRT(SUMX(VALUES('Model Calendar'[Date]); POWER([Avg Temperature] - MeanX; 2)) * SUMX(VALUES('Model Calendar'[Date]); POWER([Avg Pollution] - MeanY; 2))) RETURN DIVIDE(T; Denominator)
Climate	Corr Temperature vs Pollution	Calculates the average value for corr temperature vs pollution over the selected context.	VAR DataPoints = FILTER(ADDCOLUMNS(VALUES('Model Calendar'[Date]); -- ? IMPORTANT: use VALUES; not ALLSELECTED "Temp"; CALCULATE([Avg Temperature]); "Pollution"; CALCULATE([Avg Pollution])); NOT ISBLANK([Temp]) && NOT ISBLANK([Pollution])) VAR MeanTemp = AVERAGEX(DataPoints; [Temp]) VAR MeanPollution = AVERAGEX(DataPoints; [Pollution]) VAR Numerator = SUMX(DataPoints; ([Temp] - MeanTemp) * ([Pollution] - MeanPollution)) VAR Denominator = SQRT(SUMX(DataPoints; POWER([Temp] - MeanTemp; 2)) * SUMX(DataPoints; POWER([Pollution] - MeanPollution; 2))) RETURN IF(Denominator = 0; BLANK(); Numerator / Denominator)

AQI	Avg Station AQI	Calculates the average value for avg station aqi over the selected context.	AVERAGE (Fact_AQI[Station_AQI])
AQI	Max Station AQI	Returns the maximum value for max station aqi within the current filter context.	ROUND(MAX(Fact_AQI[Station_AQI]); 1)
AQI	Min Station AQI	Returns the minimum value for min station aqi within the current filter context.	MIN (Fact_AQI[Station_AQI])
AQI	Avg Pollutant AQI	Calculates the average value for avg pollutant aqi over the selected context.	AVERAGE (Fact_AQI[Pollutant_AQI])
AQI	Avg Pollutant Concentration	Calculates the average value for avg pollutant concentration over the selected context.	AVERAGE (Fact_AQI[Pollutant_Conc])
AQI	Worst Pollutant AQI	Returns the maximum value for worst pollutant aqi within the current filter context.	MAX (Fact_AQI[Pollutant_AQI])
AQI	Dominant Pollutant Days	Computes dominant pollutant days by modifying filter context based on the DAX expression.	CALCULATE(DISTINCTCOUNT ('Model Calendar'[Date]))
AQI	Dominant Pollutant %	Counts the number of rows to calculate dominant pollutant %.	DIVIDE(COUNTROWS (Fact_AQI); CALCULATE (COUNTROWS (Fact_AQI); ALL (Fact_AQI[Dominant_Pollutant])))
AQI	Yearly Avg AQI	Computes yearly avg aqi by modifying filter context based on the DAX expression.	CALCULATE([Avg Station AQI]; ALLEXCEPT ('Model Calendar'; 'Model Calendar'[Year]))
AQI	MoM AQI Change	Calculates the change in mom aqi change compared to a previous time period.	[Avg Station AQI] - CALCULATE([Avg Station AQI]; DATEADD ('Model Calendar'[Date]; -1; MONTH))
AQI	YoY AQI Change	Calculates the change in yoy aqi change compared to a previous time period.	[Avg Station AQI] - CALCULATE([Avg Station AQI]; SAMEPERIODLASTYEAR ('Model Calendar'[Date]))
AQI	AQI StationCategory	Computes aqi stationcategory based on the defined DAX calculation.	VAR AvgAQI = [Avg Station AQI] RETURN IF (ISBLANK (AvgAQI); "No Data"; SWITCH (TRUE(); AvgAQI <= 50; "Good"; AvgAQI <= 100; "Moderate"; AvgAQI <= 150; "Unhealthy for Sensitive"; AvgAQI <= 200; "Unhealthy"; AvgAQI <= 300; "Very Unhealthy"; "Hazardous"))
General	% Unhealthy Days	Counts the number of rows to calculate % unhealthy days.	VAR StationDays = SUMMARIZE(Fact_AQI; Fact_AQI[NAPS ID]; Fact_AQI[Date]; "AQI"; MAX(Fact_AQI[Station_AQI])) VAR UnhealthyDays = FILTER(StationDays; [AQI] > 100) RETURN DIVIDE(COUNTROWS(UnhealthyDays); COUNTROWS(StationDays))
General	% Unhealthy Days (Daily Avg)	Calculates % unhealthy days (daily avg) as a ratio between two values using safe division.	DIVIDE([Unhealthy AQI Days (Daily Avg)]; [Total Days in Period]; 0)
AQI	Unhealthy AQI Days (Daily Avg)	Counts the number of rows to calculate unhealthy aqi days (daily avg).	VAR Days = CALCULATE(COUNTROWS (VALUES ('Model Calendar'[Date])); FILTER(VALUES ('Model Calendar'[Date]); CALCULATE(AVERAGE (Fact_AQI[Station_AQI])) > 100)) RETURN COALESCE (Days; 0)
AQI	Max Dominant Pollutant Days	Returns the maximum value for max dominant pollutant days within the current filter context.	MAXX(VALUES (Fact_AQI[Dominant_Pollutant]); [Dominant Pollutant Days])
AQI	Top Pollutant	Counts the number of rows to calculate top pollutant.	VAR DailyTable = CALCULATETABLE(SUMMARIZE('Fact_AQI'; 'Fact_AQI'[NAPS ID]; 'Fact_AQI'[Date]; "DomPoll"; SELECTEDVALUE('Fact_AQI'[Dominant_Pollutant])); ALLSELECTED('Fact_AQI')) VAR CountTable = ADDCOLUMNS(SUMMARIZE(DailyTable; [DomPoll]); "Cnt"; COUNTROWS(FILTER(DailyTable; [DomPoll] = EARLIER([DomPoll])))) VAR TopRow = TOPN(1; CountTable; [Cnt]; DESC) RETURN MAXX(TopRow; [DomPoll])
AQI	AQI stationColor	Computes aqi stationcolor based on the defined DAX calculation.	VAR AvgAQI = [Avg Station AQI] RETURN SWITCH(TRUE(); AvgAQI <= 50; "#66BB6A"; -- Soft green AvgAQI <= 100; "#D4E157"; -- Muted yellow-green AvgAQI <= 150; "#FFA726"; -- Soft orange AvgAQI <= 200; "#EF5350"; -- Soft red AvgAQI <= 300; "#AB47BC"; -- Muted purple "#6D4C41" -- Brownish (hazardous))
AQI	Worst Pollutant Name	Returns the maximum value for worst pollutant name within the current filter context.	VAR MaxAQI = MAX (Fact_AQI[Pollutant_AQI]) RETURN MAXX(FILTER(Fact_AQI; Fact_AQI[Pollutant_AQI] = MaxAQI); Fact_AQI[Pollutant])
AQI	Worst AQI Date	Returns the maximum value for worst aqi date within the current filter context.	VAR MaxAQI = MAX (Fact_AQI[Pollutant_AQI]) VAR Dt = MAXX(FILTER(Fact_AQI; Fact_AQI[Pollutant_AQI] = MaxAQI); Fact_AQI[Date]) RETURN FORMAT (Dt; "dd mmm yyyy")
AQI	Worst AQI Full Info	Computes worst aqi full info based on the defined DAX calculation.	VAR Pol = [Worst Pollutant Name] VAR DateVal = [Worst AQI Date] RETURN Pol & " " & FORMAT(DateVal; "MMM YYYY")
AQI	Worst AQI Color	Computes worst aqi color based on the defined DAX calculation.	VAR AQI = [Worst Pollutant AQI] RETURN SWITCH(TRUE(); AQI <= 50; "#66BB6A"; -- Green AQI <= 100; "#D4E157"; -- Yellow AQI <= 150; "#FFA726"; -- Orange AQI <= 200; "#EF5350"; -- Red AQI <= 300; "#AB47BC"; -- Purple "#6D4C41" -- Hazardous (Brown))
AQI	Worst AQI Rank	Computes worst aqi rank based on the defined DAX calculation.	RANKX(ALLSELECTED (Dim_Stations_All[Province]); [Worst Pollutant AQI]; ; DESC)
General	Worst Provinces	Computes worst provinces based on the defined DAX calculation.	CONCATENATEX(TOPN(3; VALUES (Dim_Stations_All[Province]); [Worst Pollutant AQI]; DESC); Dim_Stations_All[Province]; ", ")
AQI	AQI Insight	Computes aqi insight based on the defined DAX calculation.	VAR AQI = [Avg Station AQI] VAR TrendVal = [YoY AQI Change] VAR Worst = [Worst Pollutant AQI] VAR Unhealthy = [% Unhealthy Days] VAR Day = [Worst AQI Date] RETURN "Overall AQI is " & IF(AQI <= 50; "GOOD ??"; "MODERATE ??") & " (Avg: " & FORMAT(AQI; "0.0") & "); " & "with " & FORMAT(Unhealthy; "0.0%") & " unhealthy days. " & "Trend is " & IF(TrendVal > 0; "worsening ??"; "improving ??") & " & "however watch for worst AQI level which reached " & FORMAT(Worst; "0") & " on " & FORMAT(Day; "dd mmm yyyy") & " ."
AQI	AQI worstpoll_ColorCode	Computes aqi worstpoll_colorcode based on the defined DAX calculation.	VAR AQI = [Worst Pollutant AQI] RETURN SWITCH(TRUE(); AQI <= 50; "Good"; AQI <= 100; "Moderate"; AQI <= 150; "Unhealthy (Sensitive)"; AQI <= 200; "Unhealthy"; AQI <= 300; "Very Unhealthy"; "Hazardous")

AQI	AQI pollutant_Color	Computes aqi pollutant_color based on the defined DAX calculation.	VAR AvgAQI = [Avg Pollutant AQI] RETURN SWITCH(TRUE(); AvgAQI <= 50; "#66BB6A"; AvgAQI <= 100; "#D4E157"; AvgAQI <= 150; "#FFA726"; AvgAQI <= 200; "#EF5350"; AvgAQI <= 300; "#AB47BC"; "#6D4C41")
AQI	Days in AQI Category	Counts the number of rows to calculate days in aqi category.	VAR MinVal = SELECTEDVALUE('AQI Legend'[Min]) VAR MaxVal = SELECTEDVALUE('AQI Legend'[Max]) VAR DailyTable = ADDCOLUMNS(SUMMARIZE(Fact_AQI; Fact_AQI[Date]); "DailyAQI"; CALCULATE(MAX(Fact_AQI[Pollutant_AQI]))) RETURN COUNTROWS(FILTER(DailyTable; [DailyAQI] >= MinVal && [DailyAQI] <= MaxVal))
AQI	Avg Daily Pollutant AQI	Calculates the average value for avg daily pollutant aqi over the selected context.	AVERAGEX(SUMMARIZE(Fact_AQI; Fact_AQI[Date]; "DailyAQI"; MAX(Fact_AQI[Pollutant_AQI])); [DailyAQI])
AQI	Map AQI max	Returns the maximum value for map aqi max within the current filter context.	MAX(Fact_AQI[Station_AQI])
AQI	Map color AQI Category	Computes map color aqi category based on the defined DAX calculation.	SWITCH(TRUE(); [Avg Station AQI] <= 50; "Good"; [Avg Station AQI] <= 100; "Moderate"; [Avg Station AQI] <= 150; "Unhealthy (Sensitive)"; [Avg Station AQI] <= 200; "Unhealthy"; [Avg Station AQI] <= 300; "Very Unhealthy"; "Hazardous")
AQI	% Most Frequent Dominant Pollutant	Counts the number of rows to calculate % most frequent dominant pollutant.	VAR TopPollutantCount = MAXX(TOPN(1; VALUES(Fact_AQI[Dominant_Pollutant]); CALCULATE(COUNTROWS(Fact_AQI)); DESC); CALCULATE(COUNTROWS(Fact_AQI))) VAR TotalCount = COUNTROWS(Fact_AQI) RETURN DIVIDE(TopPollutantCount; TotalCount)
AQI	Second Pollutant	Counts the number of rows to calculate second pollutant.	VAR DailyTable = SUMMARIZE('Fact_AQI'; 'Fact_AQI'[NAPS ID]; 'Fact_AQI'[Date]; "DomPoll"; SELECTEDVALUE('Fact_AQI'[Dominant_Pollutant])) VAR CountTable = ADDCOLUMNS(SUMMARIZE(DailyTable; [DomPoll]); "Cnt"; COUNTROWS(FILTER(DailyTable; [DomPoll] = EARLIER([DomPoll])))) VAR TopTwo = TOPN(2; CountTable; [Cnt]; DESC) VAR SecondRow = EXCEPT(TopTwo; TOPN(1; TopTwo; [Cnt]; DESC)) RETURN MAXX(SecondRow; [DomPoll])
AQI	Days > 100 (Any Pollutant)	Counts the number of rows to calculate days > 100 (any pollutant).	VAR DailyMaxTable = ADDCOLUMNS(SUMMARIZE(Fact_AQI; Fact_AQI[Date]); "MaxAQI"; CALCULATE(MAX(Fact_AQI[Pollutant_AQI]))) RETURN COUNTROWS(FILTER(DailyMaxTable; [MaxAQI] > 100))
AQI	Pollutants (>100 AQI)	Computes pollutants (>100 aqi) based on the defined DAX calculation.	VAR HighRows = FILTER(Fact_AQI; Fact_AQI[Pollutant_AQI] > 100) VAR Result = CONCATENATEX(DISTINCT(SELECTCOLUMNS(HighRows; "P"; Fact_AQI[Pollutant])); [P]; "; ") RETURN Result
AQI	Daily Station AQI	Returns the maximum value for daily station aqi within the current filter context.	CALCULATE(MAX(Fact_AQI[Pollutant_AQI]); ALLEXCEPT(Fact_AQI; Fact_AQI[NAPS ID]; Fact_AQI[Date]))
AQI	Days AQI > 50 (Station KPI)	Counts the number of rows to calculate days aqi > 50 (station kpi).	COUNTROWS(SUMMARIZE(FILTER(Fact_AQI; Fact_AQI[Station_AQI] > 50); Fact_AQI[NAPS ID]; Fact_AQI[Date]))
General	Target KPI	Computes target kpi based on the defined DAX calculation.	0.95
General	Avg Pollution	Calculates the average value for avg pollution over the selected context.	CALCULATE(AVERAGE(Fact_AQI[Pollutant_Conc]))
Wind	Avg Wind Speed	Calculates the average value for avg wind speed over the selected context.	CALCULATE(AVERAGE(Dim_wind_2015[Spd of Max Gust (km/h)]))
Trend	Trend(Deg C / Decade)	Calculates the average value for trend(deg c / decade) over the selected context.	AVERAGE(Dim_temptrend_bystation[Trend (°C / decade)])
Wildfire	Total Fires	Calculates total fires by summing the underlying values.	SUM(Dim_Wildfire[FIRES])
Wildfire	Total Area Burned	Calculates total area burned by summing the underlying values.	SUM(Dim_Wildfire[Total_Area(HA)])
Wildfire	Fires >200 HA	Calculates fires >200 ha by summing the underlying values.	CALCULATE(SUM(Dim_Wildfire[FIRES]); Dim_Wildfire[MAX_Area (HA)] > 200)
Wildfire	Avg Fire Size	Calculates avg fire size by summing the underlying values.	DIVIDE(SUM(Dim_Wildfire[Total_Area(HA)]); SUM(Dim_Wildfire[FIRES]); 0)
Wildfire	Large Fire Ratio	Calculates large fire ratio as a ratio between two values using safe division.	DIVIDE([Fires >200 HA]; [Total Fires]; 0)
Wildfire	Fire YoY Growth	Calculates fire yoy growth as a ratio between two values using safe division.	VAR PrevYear = CALCULATE([Total Fires]; SAMEPERIODLASTYEAR(Dim_Wildfire[YEAR])) RETURN DIVIDE([Total Fires] - PrevYear; PrevYear)
General	Avg Lower Bound	Calculates the average value for avg lower bound over the selected context.	AVERAGE(dim_sarima_forecast[Lower_Bound])
General	Avg Upper Bound	Calculates the average value for avg upper bound over the selected context.	AVERAGE(dim_sarima_forecast[Upper_Bound])
General	Average Actual	Calculates the average value for average actual over the selected context.	AVERAGE(Dim_temperature_actual_forecast[Actual])
General	Average Prediction	Calculates the average value for average prediction over the selected context.	AVERAGE(Dim_temperature_actual_forecast[Prediction])
General	Average LB	Calculates the average value for average lb over the selected context.	AVERAGE(Dim_temperature_actual_forecast[Lower])
General	Average UB	Calculates the average value for average ub over the selected context.	AVERAGE(Dim_temperature_actual_forecast[Upper])
General	Average Corr	Calculates the average value for average corr over the selected context.	AVERAGE('Climate&AQI_correlation'[Correlation])
Climate	Mean Temp Corr	Calculates the average value for mean temp corr over the selected context.	CALCULATE(AVERAGE('Climate&AQI_correlation'[Correlation]); 'Climate&AQI_correlation'[Weather_Variable] = "Mean Temp (°C)")
General	Avg Correlation	Calculates the average value for avg correlation over the selected context.	AVERAGE(ghg_temp_corr[GHG_Temp_Correlation])

General	Avg GHG Emissions	Calculates the average value for avg ghg emissions over the selected context.	AVERAGE(ghg_temp_corr[GHG_Emissions])
Climate	Average temp	Calculates the average value for average temp over the selected context.	AVERAGE(ghg_temp_corr[Mean Temp (°C)])